

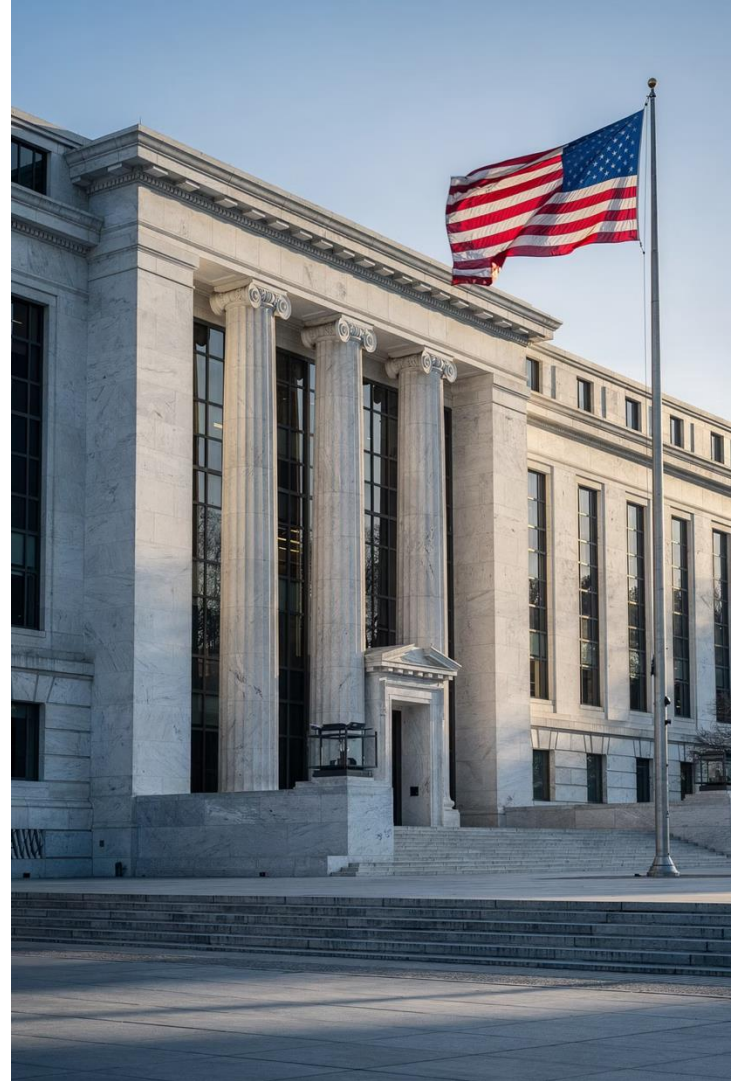


Monetary Policy

Team Teaching Financial Markets and Institutions

Monetary Policy

Understanding how the Federal Reserve shapes economic conditions through strategic control of money supply and interest rates.





CHAPTER OVERVIEW

Learning Objectives

Policy Inputs

Understand the economic indicators used to determine monetary policy decisions

Stimulative Effects

Learn how expansionary policy impacts the economy and financial markets

Restrictive Effects

Explore how contractionary policy controls inflation and spending

Policy Trade-offs

Examine the balance between unemployment and inflation objectives

Market Response

Analyze how financial markets react to Fed policy changes

Global Impact

Understand international factors affecting monetary policy decisions

The Fed's Dual Mandate



The Federal Open Market Committee (FOMC) is responsible for determining monetary policy with two primary goals: achieving a low level of inflation and a low level of unemployment. These goals are consistent with most central banks worldwide.

The Fed changes money supply growth to influence interest rates, which then affect aggregate borrowing and spending by households and firms. This spending impacts demand for products and services, ultimately affecting both price levels and unemployment.

Indicators of Economic Growth

The Fed monitors several key indicators before deciding on proper monetary policy. These indicators provide insight into the health and direction of the U.S. economy.



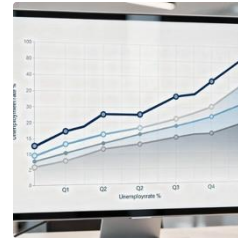
Gross Domestic Product

Measures total value of goods and services produced, serving as the most direct indicator of economic growth



National Income

Total income earned by firms and employees, reflecting demand for products and services



Unemployment Rate

Measures number of jobs being filled, though doesn't indicate job quality or discouraged workers



Sector Indexes

Industrial production, retail sales, and home sales indexes indicate growth across specific sectors



Conference Board Economic Indicators

The Conference Board publishes indexes of leading, coincident, and lagging economic indicators that are widely followed by market participants. Leading indicators predict future activity, coincident indicators move with business cycles, and lagging indicators follow economic changes.

Three consecutive monthly changes in the same direction typically suggest a turning point in the economy.

Indicators of Inflation

The Fed closely monitors price indexes and other indicators to assess the U.S. inflation rate. These measurements help predict future price movements and guide monetary policy decisions.

Producer Price Index

Reflects prices at the wholesale level, providing early signals of inflationary pressure

Consumer Price Index

Reflects prices paid by consumers at retail level, directly measuring household purchasing power

Core Inflation

Excludes volatile food and energy prices to reveal underlying inflation trends

PCE Index

Personal consumption expenditures index measures wide range of household spending





Additional Inflation Signals

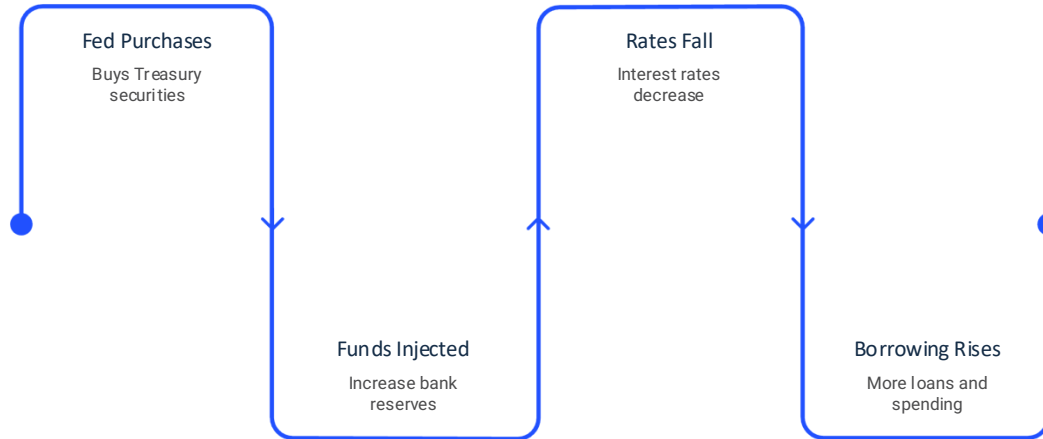
Key Indicators

Beyond traditional price indexes, several other factors signal inflationary pressure

- **Wage rates:** Highly correlated with prices over the long run
- **Oil prices:** Affect production costs, transportation, and consumer gasoline prices
- **Gold prices:** Tend to move in tandem with inflation as a hedge
- **Economic growth indicators:** Excess demand at full employment can signal demand-pull inflation

STIMULATIVE POLICY

Implementing Stimulative Monetary Policy



This streamlined process shows how the Fed stimulates economic growth through strategic market operations.

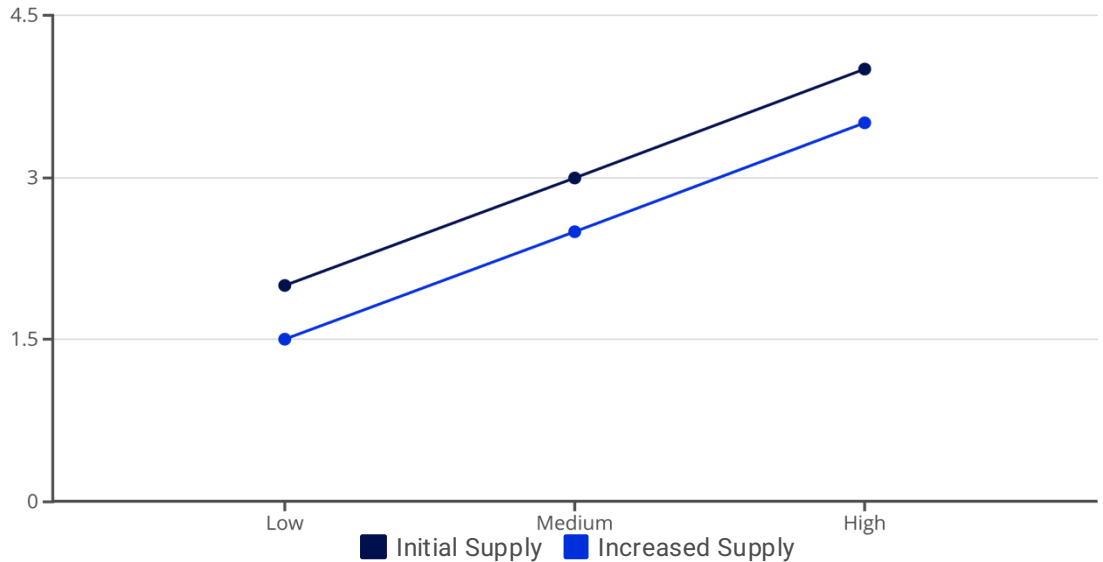
When the Fed implements a stimulative monetary policy, it uses open market operations to inject funds into the banking system to reduce interest rates. The Fed purchases Treasury securities in the secondary market, increasing account balances at financial institutions.

Lower interest rates encourage more businesses and households to borrow and spend money, which stimulates the economy and creates jobs.

How Stimulative Policy Reduces Interest Rates

When the Fed purchases Treasury securities, it increases the supply of loanable funds in the banking system. This shifts the supply curve outward, creating excess supply at the previous interest rate level.

The equilibrium interest rate declines to the level where quantities of loanable funds supplied and demanded are equal. Lower interest rates reduce the cost of financing for businesses, making more potential projects feasible.



Impact on Business Investment

Lower interest rates triggered by the Fed's stimulative monetary policy cause an increase in business investment. Firms are more willing to pursue additional projects when their cost of financing is lower.

1

Lower Rates

Fed reduces Treasury rate from 5% to 4%

2

Reduced Cost

Business borrowing cost drops from 8% to 7%

3

More Projects

Additional investments become feasible

4

Economic Growth

New spending stimulates the economy





Understanding Business Cost of Debt

Risk-Free Rate Plus Premium

The financing rate on a business loan is based on the risk-free rate (Treasury rate) plus a credit risk premium reflecting the borrower's creditworthiness.

All businesses, regardless of risk level, benefit from lower rates when the Fed implements stimulative policy.

Low Risk Business

Before: 8% (5% + 3%)

After: 7% (4% + 3%)

Moderate Risk Business

Before: 9% (5% + 4%)

After: 8% (4% + 4%)

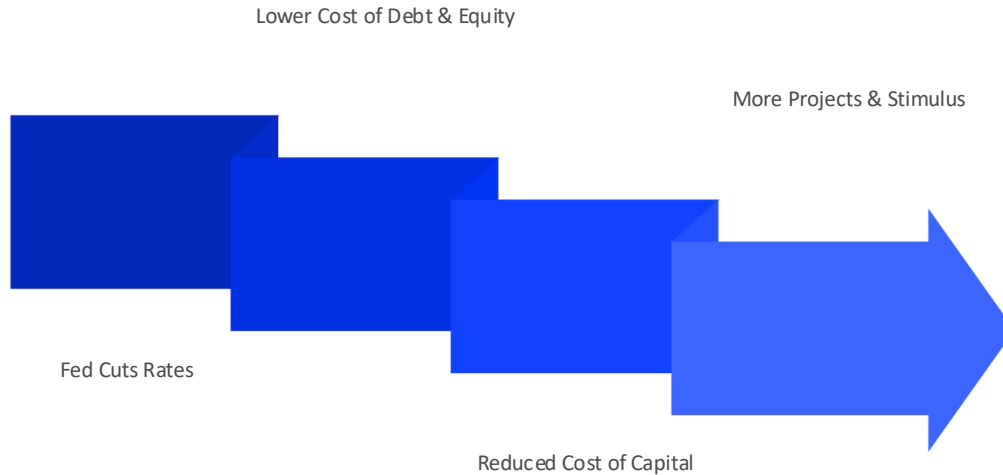
Impact on Cost of Equity



Monetary policy also influences the cost of equity. A firm's cost of equity is based on the risk-free rate plus a risk premium reflecting the firm's stock price sensitivity to market movements.

When the Fed reduces the risk-free rate by 1 percentage point, it reduces a firm's cost of equity by 1 percentage point. This lowers the firm's overall cost of capital, making more investment projects attractive.

The Complete Stimulative Effect



The Fed's ability to stimulate the economy stems from its effects on the Treasury rate, which influences both the cost of debt and cost of equity. As firms implement more projects, they spend more money and hire more employees.

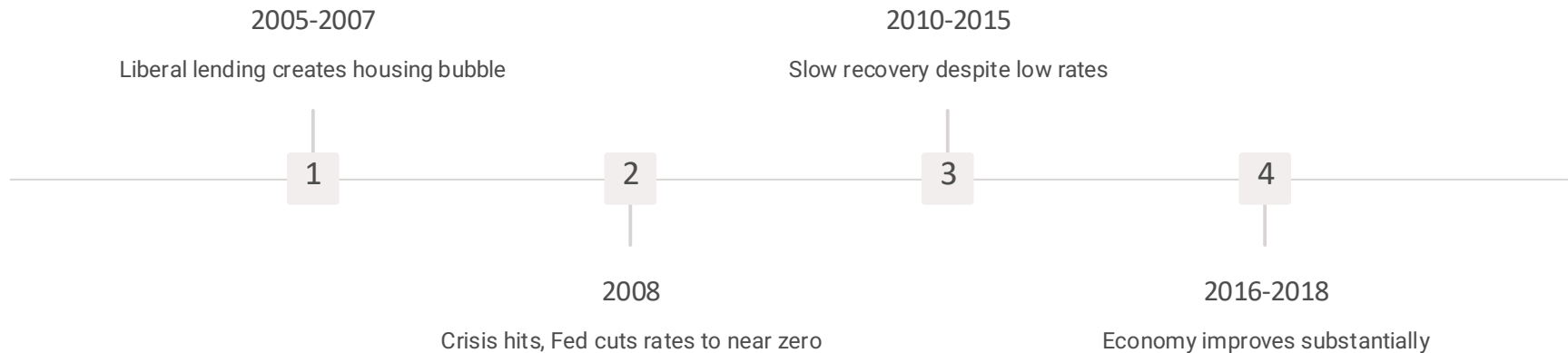
Households also benefit from lower loan rates on cars and mortgages, encouraging increased borrowing and spending that further stimulates the economy.

This comprehensive view shows how monetary policy creates a cascade of economic effects.



Case Study: The 2008 Credit Crisis

During 2005-2007, liberal lending standards led to a housing bubble. When it burst in 2008, massive layoffs occurred in construction and related industries. The Fed reduced the federal funds rate from 5.25% to near zero, but recovery was slow.





Why Stimulative Policy Might Fail

Although stimulative monetary policy is normally desirable when the economy is weak, it is not always effective. Several factors can limit its impact on economic recovery.

Limited Control of Long-Term Rates

The Fed primarily influences short-term rates, but many businesses prefer long-term fixed-rate loans

Credit Crunch

Banks may be unwilling to lend even with increased funds due to borrower uncertainty

Borrower Reluctance

Firms and households may delay borrowing until economy improves despite low rates

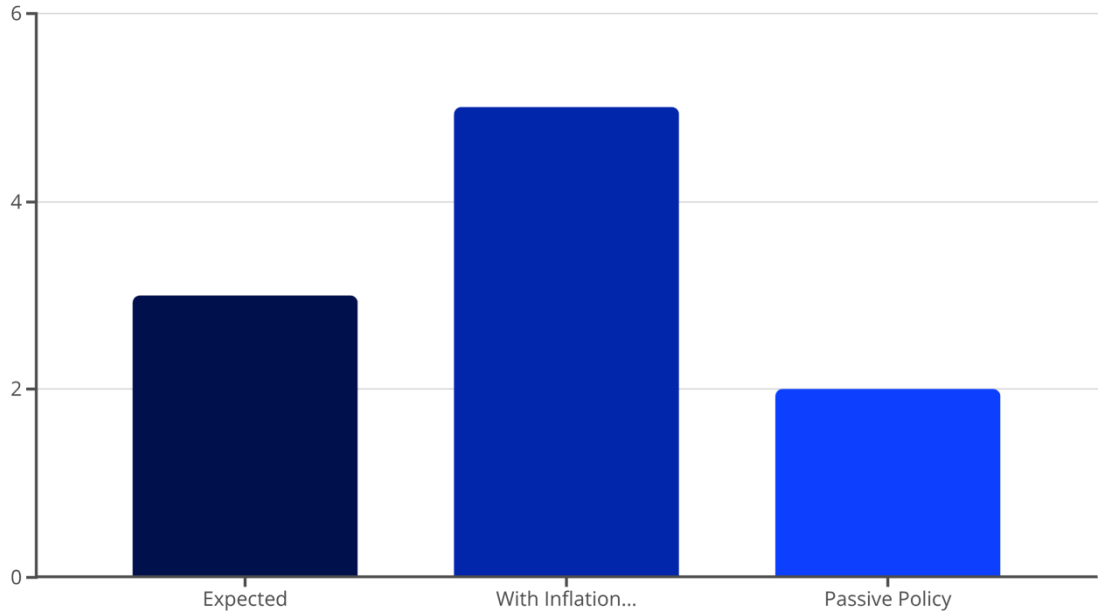
Low Returns on Savings

Near-zero interest rates reduce income for savers, potentially restricting their spending

The Rational Expectations Challenge

When the Fed increases money supply growth, businesses and households may recognize this will cause higher inflation. They revise inflationary expectations upward and rush to make purchases before prices rise.

This increases demand for loanable funds, offsetting the Fed's attempt to reduce interest rates. The result: both supply and demand increase, leaving interest rates unchanged and limiting the policy's effectiveness.





Monetary Policy Lags

The Challenge of Timing

Three lags make the Fed's job more challenging. By the time a policy takes full effect, economic conditions may have changed, potentially making the policy inappropriate or even harmful.

For example, a stimulative policy implemented during high unemployment might not take effect until unemployment has already declined, potentially causing the economy to overheat and creating inflation.

RESTRICTIVE POLICY

Implementing Restrictive Monetary Policy



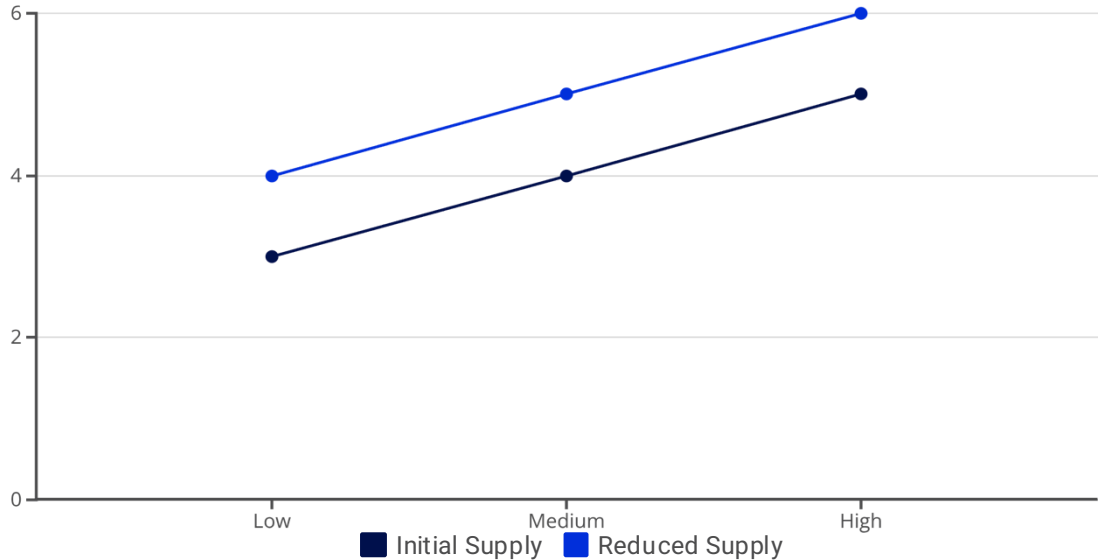
When excessive inflation is the main concern, the Fed implements a restrictive (tight-money) policy by reducing money supply growth. This approach targets demand-pull inflation caused by excessive economic growth.

The Fed sells Treasury securities in the secondary market. As investors purchase these securities, their account balances decrease, resulting in a net decrease in loanable funds and upward pressure on interest rates.

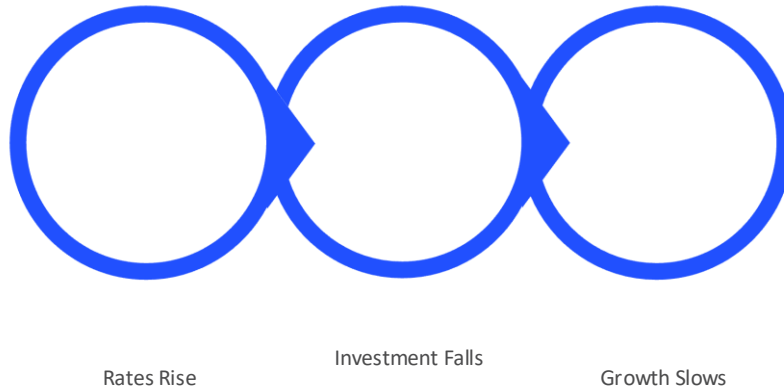
How Restrictive Policy Raises Interest Rates

When the Fed sells Treasury securities, the supply of loanable funds decreases. This shifts the supply curve inward, creating excess demand at the previous interest rate level.

Depository institutions raise rates on federal funds market loans, deposits, and household and business loans. The equilibrium interest rate increases to the level where quantities supplied and demanded are equal.



Impact on Business Investment and Inflation



Higher rates cool the economy by making borrowing more expensive for businesses and households.

Higher interest rates increase the corporate cost of financing new projects, causing a decrease in business investment. As economic growth slows, inflationary pressure is reduced.

Example: If the Fed raises the Treasury rate from 5% to 6%, a firm paying a 4% risk premium must now pay 10% to borrow funds, making fewer projects economically viable.



Comparing Monetary Policy Approaches

Stimulative Policy

Action: Fed buys Treasury securities

Effect: Bank funds increase, interest rates decrease

Result: Aggregate spending and economic growth increase

Restrictive Policy

Action: Fed sells Treasury securities

Effect: Bank funds decrease, interest rates increase

Result: Aggregate spending decreases, inflation reduces



The Inflation-Unemployment Trade-off

The Fed faces a fundamental challenge: it's difficult to achieve both very low unemployment and very low inflation simultaneously. Strong economic conditions typically bring low unemployment but higher inflation, while weak conditions bring lower inflation but higher unemployment.

The Fed must carefully balance these competing objectives when determining monetary policy.

External Forces Affecting the Trade-off



Other forces beyond the Fed's control can affect the inflation-unemployment trade-off. Energy costs, structural unemployment, and global economic conditions can shift the entire range of possible outcomes.

For example, if oil prices increase substantially, the minimum inflation rate rises regardless of monetary policy. Similarly, if training centers close, the minimum unemployment rate increases. These external factors constrain the Fed's ability to achieve its dual mandate.



Monetary Policy and Fiscal Policy Interaction

The Fed's monetary policy is influenced by the government's fiscal policies. Large budget deficits can place upward pressure on interest rates, potentially dampening the economy.

The Fed may feel pressured to use stimulative monetary policy to offset the effects of fiscal policy. This interaction between monetary and fiscal policy adds complexity to the Fed's decision-making process.

Fiscal Policy Impact

Budget deficits increase demand for funds, raising interest rates

Fed Response

May implement stimulative policy to counteract higher rates

Coordination Challenge

Fed must balance its goals with fiscal policy effects



The Inflation Targeting Debate

Some economists propose that the Fed should focus primarily on controlling inflation rather than balancing inflation and unemployment objectives. This approach, called inflation targeting, could make the Fed's role more transparent.

However, critics argue this could result in higher unemployment and that the Fed might lose credibility if inflation deviates from targets due to factors beyond its control.

Impact on Financial Markets

Monetary policy has a profound influence on financial markets because it affects interest rates and economic growth, which in turn affect the valuation of most securities.



Bond Markets

Bond values move inversely with interest rates.

Investors benefit when the Fed lowers rates but face losses when rates rise.



Stock Markets

Stock values are affected by interest rate changes, but effects are less consistent than for bonds. Lower rates may boost stocks if they signal growth.



Mortgage Markets

Mortgage rates closely follow Fed policy. Lower rates increase housing demand while higher rates cool the housing market.

Quantitative Easing: 2008-2014

During the credit crisis, the Fed implemented quantitative easing—a unique strategy of purchasing specific dollar amounts of risky debt securities each month, not just Treasury securities.

By purchasing mortgage-backed securities and commercial paper, the Fed attempted to boost investor confidence and increase liquidity in markets for these securities when traditional monetary policy proved insufficient.



FOMC Communication to Markets



After each FOMC meeting, the committee issues a statement explaining its monetary policy decision. These statements have significant implications for security prices.

"The Federal Open Market Committee decided to reduce its target for the federal funds rate by 0.25%. Economic growth has weakened, and indicators suggest more pronounced weakness. The Committee expects weakness will continue."

Financial market participants closely analyze these statements to interpret the Fed's future plans and assess impacts on security prices.



Global Monetary Policy Considerations

The Fed's monetary policy decisions are influenced by international factors. Financial markets are globally integrated, requiring the Fed to consider worldwide economic conditions.



Dollar Impact

A weak dollar stimulates U.S. exports and reduces import competition. A strong dollar dampens the economy but reduces inflation.



Interest Rate Transmission

U.S. interest rate changes attract or repel foreign investment, affecting global capital flows and interest rates worldwide.



Global Conditions

Strong foreign economies increase demand for U.S. products. Weak global conditions require more aggressive Fed intervention.



Global Crowding Out

U.S. budget deficits can raise interest rates globally as foreign investors shift funds to U.S. securities.



Key Takeaways

Policy Tools

The Fed uses open market operations to adjust money supply, influencing interest rates and economic activity

Dual Objectives

Balancing low inflation and low unemployment requires careful trade-offs and consideration of economic indicators

Market Impact

Monetary policy significantly affects financial markets, influencing bond values, stock prices, and mortgage rates

Global Integration

International factors including currency values, foreign economic conditions, and global capital flows shape Fed decisions

Policy Limitations

Recognition, implementation, and impact lags, along with external forces, can limit monetary policy effectiveness

Ongoing Evolution

The Fed continues to adapt its approach, including innovative strategies like quantitative easing during crises



Thank you